

SIREN: Self-supervised 4D Implicit Representation for Enhanced Radar- and Vision-based Activity Recognition for Ship Watch-Keeping Officers

Mengchuang Hu, Mozi Chen*, Yiling Ren, Junjie Weng, Xuedou Xiao, Shengkai Zhang and Kezhong Liu

Abstract—Ensuring the safety of ship operations requires accurately and consistently identifying and alerting unsafe activities of watch-keeping officers under all conditions. However, current methods, which primarily rely on wearable devices and single-modal sensors like smart bands and cameras, often fall short in complex shipboard environments due to challenges such as fluctuating lighting, varying officer positions, and frequent obstructions. To address these limitations, this paper proposes a 4D implicit representation-based multimodal activity recognition method that integrates data from both cameras and mmWave radar, leveraging recent advances in vision computing. Specifically, to bridge the spatial resolution gap between mmWave radar and cameras, we first propose a self-supervised learning framework that enhances radar angular resolution. This framework allows a single-chip radar to emulate the signal quality of a cascaded radar system, achieving a fourfold improvement in spatial resolution. Furthermore, we employ an occupancy flow model to effectively learn representations of ship officers' activities across visual and radar modalities. This neural network captures the 3D spatial and 1D temporal dynamics of human activities, providing a deeper understanding of movement over time and space. Additionally, we use contrastive learning to mitigate cross-modal conflicts during training. To support the proposed method, we defined key activities on ship watch-keeping officers and created relevant datasets in both real and simulated ship environments.¹ Experimental results demonstrate the superiority of our approach over traditional methods, with our model achieving precision and recall values of 88.29% and 86.58% in normal cabin environments, and 85.21% and 83.20% in low-light cabin environments, respectively, highlighting its potential to significantly improve maritime safety.

Index Terms—Ship Bridge Officers Activity Recognition, Radar Resolution Enhancement, Radar-Vision Fusion, Implicit Representation.

I. INTRODUCTION

Recognition and alerting of ship watch-keeping officers' unsafe activities is essential for ensuring maritime safety. According to the European Maritime Safety Agency, human factors have been implicated in over 80.7% of marine accidents in the past decade [1]. A robust and intelligent system for recognizing risk-related activities holds great potential for enhancing ship safety.

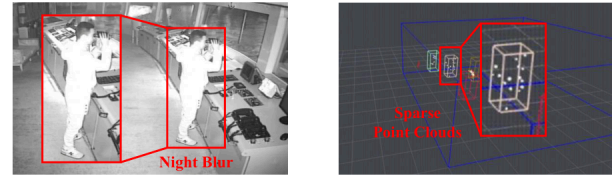
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¹https://github.com/xfuhmc/SIREN_Dataset



(a) Visual Sensors at Night.

(b) Radar Point Clouds.

Fig. 1: Limitations of visual and radar sensors. Visual sensors are prone to issues of image blur and occlusion at night. Radar point clouds suffer from low spatial resolution.

The existing driving behavior recognition techniques can be roughly classified into two categories, i.e., *device-based* [2]–[6] and *device-free* [7]–[13] methods. *Device-based* methods involve the use of wearable sensors attached to the driver's body to collect vital signs [2], [3] and activity data [4]–[6]. These sensors include smartwatches [5], accelerometers [4], electroencephalograms (EEG) [2], [3], and radio frequency identification (RFID) devices [6]. While these techniques achieve high accuracy in detecting human movements through micro-IMU sensors, they are constrained by device wearing requirements and limited battery life.

On the other hand, *device-free* methods, which mainly rely on pre-installed cameras and mmWave radar, enable more seamless and long-term monitoring. These methods provide accurate and fine-grained sensing results through advances in computer vision (CV) and artificial intelligence (AI). CV-based techniques include eye tracking [7], head pose estimation [8], hand and body movement analysis [9], utilizing AI tools such as YOLO to detect drowsiness, distraction, or inattention [10]. Radar-based methods use millimeter-wave radio signals to recognize behaviors by detecting motion parameters [13] and monitoring vital signs like heartbeats [11] and breathing rate [12]. As shown in Figs. 1 and 2, CV-based methods achieve high accuracy but are sensitive to lighting conditions and occlusions, whereas radar works well in low-light conditions but suffers from low resolution. Given these complementary characteristics, combining these techniques is well-suited for ship watch-keeping officer activity recognition, especially in complex ship bridge environments with large spaces and numerous obstructions, which make single-modality approaches impractical.

Currently, multimodal activity recognition methods can be grouped into two primary categories, i.e., *aggregation-*

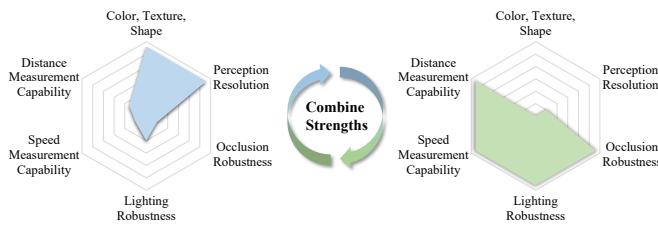


Fig. 2: Strengths of visual and radar sensors. Visual sensors offer high spatial resolution. Radar sensors are immune to lighting effects.

based [14]–[16] and *alignment-based* [17]–[20] approaches, as illustrated in Fig. 3. *Aggregation-based* approaches combine the outputs of different modality-specific subnetworks using operations like addition, concatenation, or attention mechanisms to form a joint representation. While this approach facilitates the identification and enhancement of similar information, it may overlook important modality-specific details. Therefore, this method is primarily used for information fusion between similar modalities or intra-modalities, such as RGB and depth data [15], or both the color features and the skeleton features extracted from RGB [14], [16]. Conversely, *alignment-based* methods focus on aligning the embeddings of different modalities through regularization losses to form coordinated representations, ensuring that they represent similar information. However, these methods typically fall short of facilitating the exchange of information within each modality, limiting the potential for cross-modal interaction. It is mainly applied for cross-modal alignment, but most methods independently sum the decision scores of each modality at the final stage [17], [18]. Table I summarizes these methods and their limitations. In contrast to these traditional methods, the emergence of **implicit representations**, such as occupancy network (ONet) [21] and NeRF models [22] offers a promising new approach. By employing high-dimensional functions f_θ to model the surface of objects, implicit representations can provide continuous and arbitrarily high-resolution spatial and temporal data. This capability makes them more effective than conventional explicit representations in capturing intricate details and dynamics. Our insight is to utilize a shared neural network to implicitly represent f_θ across two different modalities, enabling the joint modeling of 4D spatiotemporal dynamics from both radar and visual data. Through iterative learning on data from these modalities, we can refine and optimize the representation, effectively addressing the limitations inherent in traditional *aggregation-based* and *alignment-based* fusion methods.

However, applying implicit representations to activity recognition presents unique challenges. Firstly, although radar is more robust than vision when depth information is crucial, it has low pitch/roll resolution due to a limited antenna array,

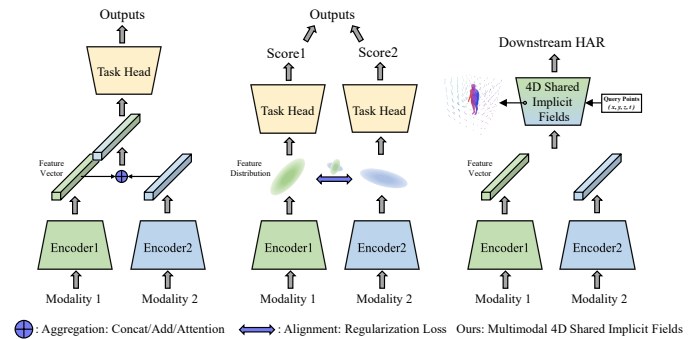


Fig. 3: Comparison of aggregation-based alignment-based and our method. Our method learns a multimodal 4D shared implicit field to jointly model human spatiotemporal dynamics from radar and visual data.

which can lead to suboptimal fusion results [23]. Secondly, existing multimodal fusion methods often fall short when it comes to preserving intra-modal details or fully exploiting cross-modal interactions [14]–[18], [24]. Thirdly, existing datasets are primarily focused on land-based driving environments, with limited data available for ship cabin scenarios.

In this paper, we present SIREN, a novel 4D implicit representation activity recognition framework. First, to overcome the problem of the resolution gap between the radar and vision, we introduce the Radar Enhancement Module, achieving a fourfold improvement of its angular resolution, which enables the radar to capture more detailed spatial information for effective multimodal fusion with camera data. Second, to overcome the problem of the current multimodal learning methods' limitation, we propose using 4D implicit representations as constraints within a shared neural network to facilitate cross-modal learning, capturing complementary features from both radar and visual inputs, we use contrastive learning to address the modality interference and reduce feature misalignment between modalities. Third, to address the scarcity of datasets for ship watch-keeping officer activity recognition, we collected a dataset specifically for ship cabin scenarios and compared it with mainstream activity recognition methods to evaluate the effectiveness of our method. Experimental results demonstrate that our approach outperforms traditional methods in complex ship cabin scenarios. In summary, our contributions are as follows:

- 1) We have collected and open-sourced a dataset for ship watch-keeping officer activity recognition, which includes RGB images and radar signals. The dataset consists of five key activities and a total of 8701 clips.
- 2) We employ 4D implicit representations as constraints, enabling cross-modal learning to overcome challenges faced by traditional multimodal fusion techniques.
- 3) We propose a self-supervised learning framework that enhances the radar angular resolution by 4x, addressing the limitations of conventional radar systems with low angular resolution.

TABLE I: Comparison of existing activity recognition methods in terms of modality, strengths, and limitations.

Category	Method	Modality	Strength	Limitation
Single-Modality	Vision-based [7]–[10]	RGB Images / Videos	Fine-grained spatial detail	Sensitive to lighting conditions and occlusions
	Radar-based [11]–[13]	mmWave Radar Signals	Robust to illumination variations	Limited spatial resolution
	Device-based [2]–[6]	Wearable Sensor Signals	Accurate motion and physiological sensing	Constrained by wearing requirements and battery life
Multimodal Fusion	Aggregation-based [14]–[16]	RGB–Depth / RGB–Skeleton	Effective for fusing and enhancing similar intra-modal features	May overlook modality-specific details
	Alignment-based [17]–[20]	Skeleton–Text / Heterogeneous Signals	Effective for improving cross-modal consistency	Limited information exchange within each modality
Ours	SIREN	Vision–Radar	4D shared implicit field enabling cross-modal spatiotemporal modeling	Mitigates limitations of existing methods

II. RELATED WORK & PRELIMINARY

Human activity recognition (HAR) involves identifying and understanding human actions using sensors and AI techniques, applicable to various tasks such as elder healthcare, smart home and driving behavior monitoring. We briefly review HAR from two aspects, i.e., different sensors and different human representations. In addition, we will introduce a new form of representation for HAR, i.e., the Occupancy Flow representation [25].

A. Sensing Modalities for Human Activity Recognition

The sensing techniques for human activity recognition can be classified into three categories: computer vision (CV), radio frequency (RF) and wearable devices. Wearable device-based methods attach sensors (e.g., IMUs, smartwatches) to specific body parts for movement monitoring. These methods [26], [27] achieve high accuracy through micro-sensors and vital sign monitoring, but their practical applications are limited by device wearing requirements and battery constraints.

CV-based methods capture rich textural information through techniques like eye tracking [7], head pose estimation [8], and body movement analysis [9]. While advanced networks [28] attempt to handle view changes and partial occlusions through view transformation, they remain sensitive to environmental factors like illumination variations. RF-based methods demonstrate robustness by penetrating occlusions to extract 3D location, Doppler velocity [29], and vital signs [11]. Although recent works enhance radar data quality using cross-section intensities [30] and pre-trained models [31], spatial resolution remains a key limitation.

Given these complementary characteristics, with CV's high spatial resolution but environmental sensitivity, and radar's robustness but limited resolution, fusing these modalities is promising, yet challenging because the key lies in how cross-modal interaction and consistency are enforced. Recent works have explored this from different angles, including hierarchical multi-modal knowledge modeling for action understanding [32], prototype-guided cross-modal alignment to reduce distribution mismatch [19], cross-modal federated representation learning that disentangles modality-shared and modality-specific features [20], and perception-aware cross-modal fusion between heterogeneous sensors [33].

Although these methods differ in architectural design and learning paradigms, they all operate at feature-level fusion or embedding-level alignment, which is often insufficient to ensure spatiotemporal consistent interaction between vision and low-resolution mmWave radar under challenging conditions such as illumination variations and occlusions. This suggests the need for a representation that can more explicitly capture shared spatiotemporal structure across heterogeneous modalities.

B. Human Behavior Representations

After sensor data processing, the input data can be converted into different intermediate representations such as skeleton [29], [34], point cloud [35], [36], and mesh [37], [38]. These representations enhance model generalization to new environments and individuals [29]. The common human representations are shown in Fig. 4.

Skeleton representation uses key points to represent joints of a human body, providing basic structure and pose information with vertex-level correspondence for temporal features. [34] uses graph convolution networks to capture spatiotemporal relationships in skeleton data, while [29] employs attention frameworks for temporal information extraction. However, skeletons lack fine-grained details, making them unsuitable for complex scenarios like object interactions. While [14], [16] attempt to combine skeleton data with image features, they often fail to capture fine details across different scenarios.

Point cloud and mesh representations offer more detailed spatial modeling. Point cloud provide rich geometric information but suffer from unstructured distribution, while meshes add surface continuity through polygonal connections but are sensitive to noise and incomplete data. Both face challenges in temporal modeling due to their unordered nature. Recent works address these limitations through voxelization [35], attention-based connectivity [38], and trajectory modeling [39], yet achieving robust spatiotemporal correspondence remains an open challenge.

Our insight is to utilize a new form of representation for the HAR task: Occupancy Flow (OFlow) representation [25], which employs nonlinear functions to model continuous object surfaces in both spatial and temporal dimensions. This approach provides finer spatial details than traditional represen-

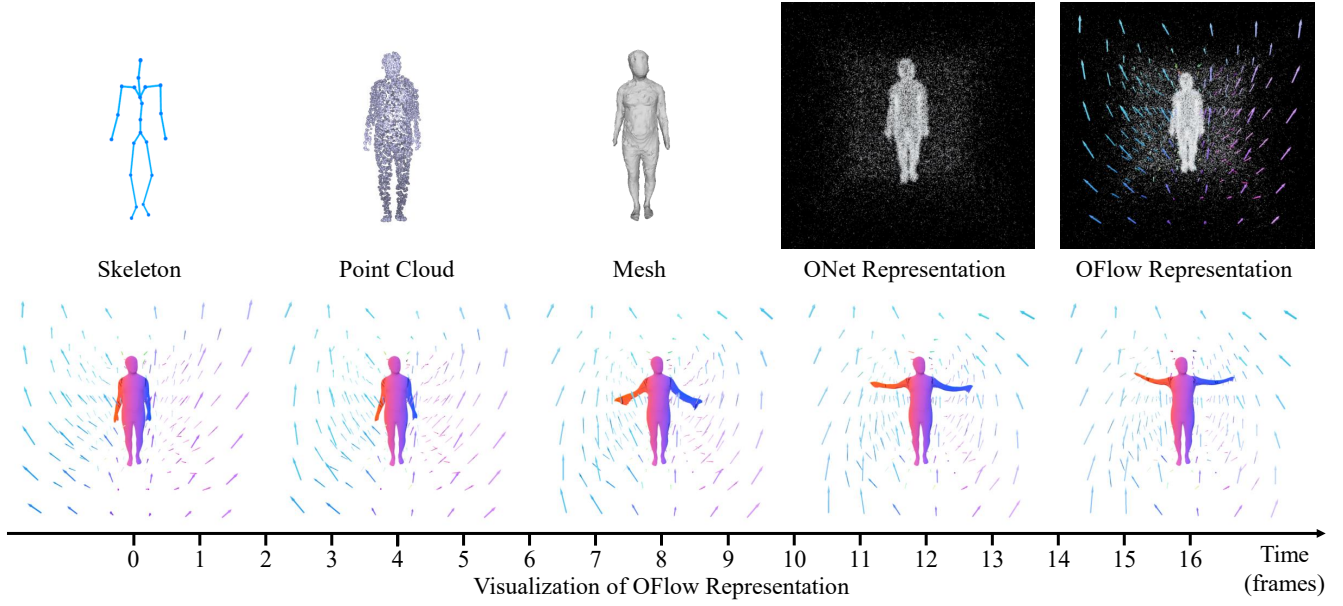


Fig. 4: Overview of existing human representations. Skeleton representation uses key points and lines to depict the body's pose, but lacks fine details. Point cloud representation uses points to represent the body, but the points are disordered. Mesh representation builds on this by connecting the points with polygons, but is prone to holes and artifacts. The ONet representation [21] uses neural networks to describe objects using probability distributions, it has continuity and supports arbitrary resolutions with the properties of neural networks. The OFlow representation [25] extends this by adding a temporal dimension, establishing spatiotemporal correspondence.

tations while maintaining temporal correspondence, enabling more accurate capture of human activity dynamics.

C. Occupancy Flow Representation

Occupancy network (ONet) [21] is a type of implicit representation, which takes an observation $x \in \mathcal{X}$ and any point $p \in \mathbb{R}^3$ as inputs, outputting the occupancy probability of the point. The core concept lies in its ability to represent 3D geometry implicitly by modeling the object's surface using a continuous function f_θ :

$$f_\theta : \mathbb{R}^3 \times \mathcal{X} \rightarrow [0, 1] \quad (1)$$

where θ are the network parameters. This function defines 3D surfaces through the continuous decision boundary of a neural network classifier, indicating whether a point resides inside or outside the object's surface.

Based on ONet, OFlow [25] extends temporal modeling through a velocity network $v_{\hat{\theta}}$ that learns a deformation gradient field:

$$v_{\hat{\theta}} : \mathbb{R}^3 \times [0, T] \rightarrow \mathbb{R}^3 \quad (2)$$

with the constraint relationship between speed $v(p, t)$ and trajectory $s(p, t)$:

$$\frac{\partial s}{\partial t}(p, t) = v(s(p, t), t) \quad \text{s.t.} \quad s(p, 0) = p \quad (3)$$

Using Neural ODEs [40], we can obtain continuous trajectories of points over time. This representation has been successfully applied in 3D model reconstruction [41], allowing

for detailed local deformations and comprehensive modeling of human body parts.

However, no existing work combines this implicit representation with multimodal fusion for HAR. Our insight is that the 4D implicit representation offers significant advantages in both spatial and temporal dimensions, making it highly suitable for improving HAR generalization and accuracy. We propose enhancing this representation through multimodal fusion to effectively address illumination variation and occlusion challenges in HAR.

III. SYSTEM DESIGN

In this section, we first provide an overview of SIREN, followed by the Radar Enhancement Module to improve the radar's angular resolution. Next, we describe the Implicit Fusion Module, which builds a human 4D implicit representation from multimodal data. Finally, we explain how the activity recognition is obtained.

A. Overview

The architecture of our model is shown in Fig. 5, comprising three main stages: Stage 1 for radar enhancement, Stage 2 for implicit multimodal fusion, and Stage 3 for activity recognition.

- **Multimodal Data Capture:** First, multimodal data from the camera and mmWave radar sensors are collected, producing a 2D image sequence and radar raw data, respectively.

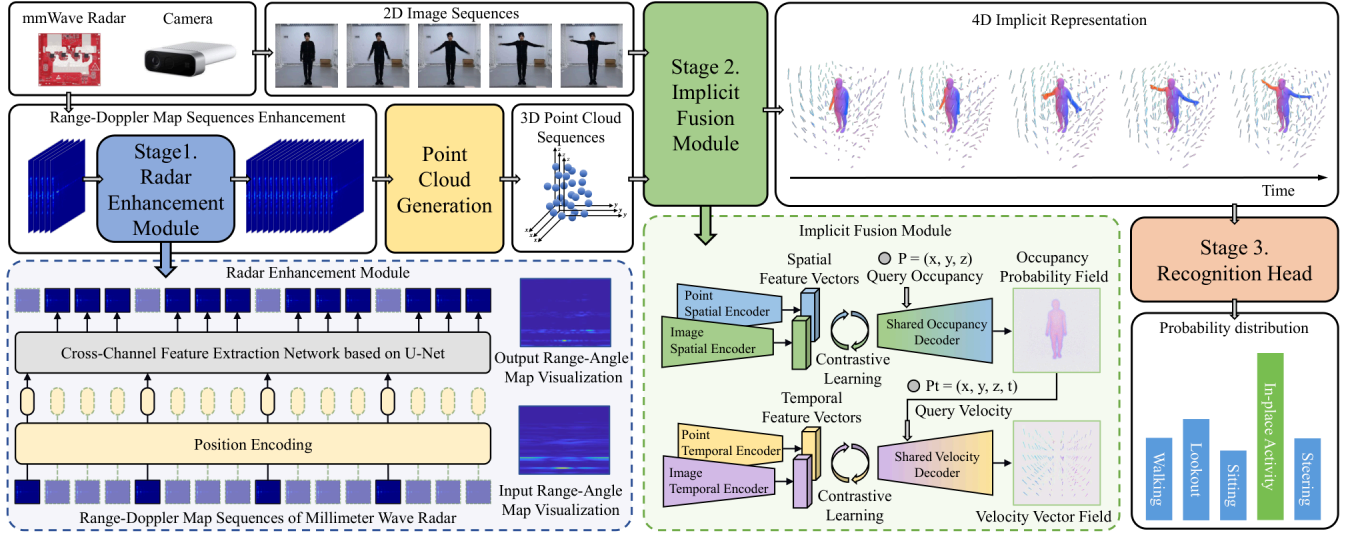


Fig. 5: Overview of the proposed SIREN framework, consisting of three stages. **Stage 1: Radar Enhancement Module** improves the angular resolution of mmWave radar signals by reconstructing dense range-Doppler maps and generating higher-quality 3D point cloud sequences. **Stage 2: Implicit Fusion Module** jointly encodes radar point clouds and image sequences into a shared 4D implicit representation, modeling both spatial occupancy and temporal motion through cross-modal learning. **Stage 3: Recognition Head** leverages the learned 4D representation to predict activity labels via a PointNet-based classifier.

TABLE II: Pipeline summary of the SIREN framework.

Module	Input	Output	Training Objective
Radar Enhancement Module	Raw 4-channel range-Doppler maps	Enhanced 16-channel range-Doppler maps	Self-supervised masked channel reconstruction with MSE loss
Implicit Fusion Module	Image sequences and point cloud sequences	4D mesh sequence reconstructed from learned implicit occupancy and velocity fields	BCE reconstruction loss and contrastive learning loss
Recognition Head	Vertex subsequences extracted from the reconstructed 4D mesh sequence	Activity probability distribution	Cross entropy classification loss

- **Radar Enhancement Module:** To bridge the resolution gap between radar and vision, the radar range-Doppler map sequences are passed through the Radar Enhancement Module, which enhances the resolution by 4x improvement. We then transfer the map sequences into 3D point cloud sequences.
- **Implicit Fusion Module:** Next, the 2D image sequences and 3D point cloud sequences are fused by the Implicit Fusion Module, which extracts spatial and temporal features from both modalities. Using a shared occupancy decoder and a shared velocity decoder, the module predicts the occupancy probability field and the velocity vector field, establishing a 4D human implicit representation that is correlated in both space and time.
- **Activity Recognition:** Leveraging the spatiotemporal correspondence of the 4D implicit representation, we extract the corresponding vertices and use a 3-layer PointNet [42] to obtain the final activity recognition results.

The overall pipeline of the proposed SIREN framework is summarized in Table II.

B. Radar Enhancement Module

In this part, we introduce the Radar Enhancement Module to bridge the resolution gap between the radar and vision. A brief

diagram is shown in Fig. 6. The module is trained on large-scale radar by masking channels and using self-supervised learning. It enhances angular resolution by 4x when applied to small-scale radar during inference. We will discuss the Cascaded Radar vs. Single Chip Radar, Module Architecture, Training Process, and Transfer Process.

Cascaded Radar vs. Single-Chip Radar. We briefly introduce the comparative analysis of the two radar types, summarized in Table III. Our training data is derived from both cascaded radar and single-chip radar. The effectiveness of our proposed framework stems from the fact that the cascaded radar and single-chip radar share similar hardware designs, with the primary difference being the antenna array size. This design allows our module achieves the same angular resolution as a cascaded radar while utilizing a smaller-sized radar at a lower cost.

Module Architecture. To achieve the above goal, the architecture of the proposed module is illustrated in Fig. 7. It primarily consists of two components: a positional encoding function and a radar cross-channel feature extraction network based on U-Net [43]. U-Net is chosen for its ability to effectively capture spatial features through its encoder-decoder architecture, making it well-suited for cross-channel feature extraction in radar data.

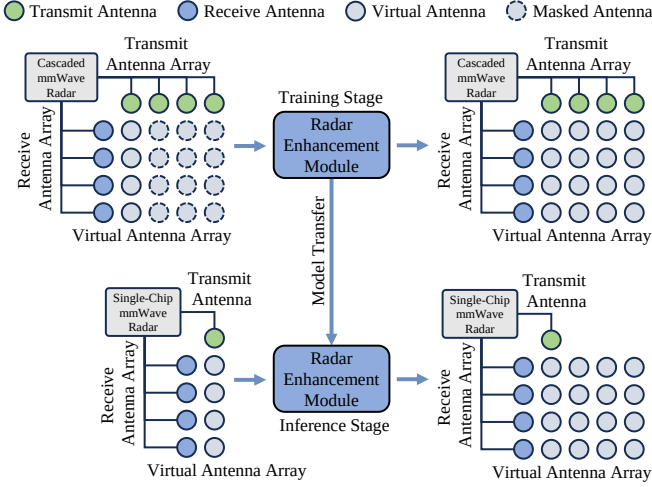


Fig. 6: Diagram of the Radar Enhancement Module, illustrating the training and inference stages. In the training stage, specific channels of a cascaded radar are masked, and self-supervised learning is employed to predict the masked channels. In the inference stage, the trained module is deployed on a single-chip radar.

TABLE III: Millimeter Wave Radar parameter configuration

Radar Model	IWR6843ISK	MMWCAS-RF-EVM
ADC Sampling Rate	128	128
Number of Chirps per Frame	64	64
Number of Transmit Antennas	3	12
Number of Receive Antennas	4	16
Price	US\$175.00	US\$1,099.00

For model input, a 12×16 range-Doppler matrix is firstly collected using cascaded radar data. The transmitting channel matrix is superimposed with the corresponding receiving channels to form a 16-channel radar range-Doppler matrix. We select channels 1st, 5th, 9th, and 13th as input and mask the remaining channels (for different masking strategies, see ablation studies). Such a sparse uniform distributed selection increases the effective array aperture L_{eff} , which directly determines the angular resolution $\Delta\theta$ of the radar array according to:

$$\Delta\theta \approx \frac{\lambda}{L_{\text{eff}}} = \frac{\lambda}{(x_{\text{max}} - x_{\text{min}})}, \quad (4)$$

where λ is the radar wavelength and $x_{\text{max}}, x_{\text{min}}$ denote the positions of the farthest selected antenna elements along the array axis. Then a position encoding function (Equation 5) is applied to the input channels, as the relative position information between channels is crucial to maintaining coherence and accurately reconstructing the radar array response.

$$pe(pos, i) = \begin{cases} \sin\left(pos \cdot \frac{1}{10000^{2i/d}}\right), & \text{if } i \text{ is even} \\ \cos\left(pos \cdot \frac{1}{10000^{2i/d}}\right), & \text{if } i \text{ is odd} \end{cases} \quad (5)$$

where $pe(pos, i)$ represents the position vector, pos denotes the position in the sequence of complex matrices for each channel, i represents the index corresponding to the complex matrix, and d represents the dimensionality of the input matrix.

Based on that, we use a U-Net-based network to learn the features and predict the remaining masked channels. The input channels are then combined with the predicted channels to obtain the final result.

Training Process. During training, a self-supervised learning strategy is employed. Specifically, we mask all radar channels except for the 1st, 5th, 9th, and 13th channels, and the module is tasked with predicting the masked data using the remaining channels. A mean squared error loss function is used to measure the difference between the predicted results and the ground truth:

$$\mathcal{L}_{rd} = \frac{1}{N_i N_j} \sum_{i,j} \left(y_{i,j}^{\text{pred}} - y_{i,j}^{\text{label}} \right)^2 \quad (6)$$

where N_i is the number of samples, and N_j represents the number of channels in a sample. $y_{i,j}^{\text{pred}}$ is the predicted Range-Doppler matrix for the j -th virtual channel in the i -th sample.

Transfer Process. Once the Radar Enhancement Module is trained using data from the cascaded radar system, the learned parameters are transferred to a single-chip mmWave radar system. This enables the smaller radar system to achieve enhanced angular resolution, similar to the cascaded radar system. By leveraging the intrinsic patterns learned from the large-scale radar system, the single-chip radar can reconstruct high-resolution angular data from fewer virtual channels. We visualize the input and output range-angle map as shown in Fig 5.

C. Implicit Fusion Module

In this part, we introduce the Implicit Fusion Module to address the limitations of traditional fusion methods. It is proposed based on the implicit representation OFlow [25]. Using OFlow representation as a constraint, the module learns from both radar point clouds and visual modalities, capturing the spatial and temporal features of the human body, and establishing a 4D spatiotemporal correspondence. We will discuss the following components: Cross-Modal Encoder, Shared Decoder, Contrastive Training, and 4D Representation Inference.

Cross-Modal Encoder. Given an image sequence $\mathbf{I} \in \mathbb{R}^{H \times W \times T \times 3}$ (where H and W represent the height and width of each image, T is the number of time steps, and 3 indicates the RGB channels) and a point cloud sequence $\mathbf{P} \in \mathbb{R}^{N_p \times T \times 3}$ (where N_p is the number of points per frame, and T is the number of time steps), we apply two parallel ResNet based PointNet to encode the first frame and the full point cloud sequence, extracting spatial and temporal features. Specifically:

$$f_{\text{PointNet_spatial}}(\mathbf{P}_{t=0}) = \mathbf{c}_{\text{radar}}^{\text{spatial}} \in \mathbb{R}^{512} \quad (7)$$

$$f_{\text{PointNet_temporal}}(\mathbf{P}) = \mathbf{c}_{\text{radar}}^{\text{temporal}} \in \mathbb{R}^{512} \quad (8)$$

where $\mathbf{P}_{t=0}$ is the first frame, and $\mathbf{P}$ is the entire point cloud sequence.

Similarly, the visual branch extracts spatial and temporal features using ResNet-18 [44] and a 3D convolutional network (the detailed architecture is shown in Fig. 8):

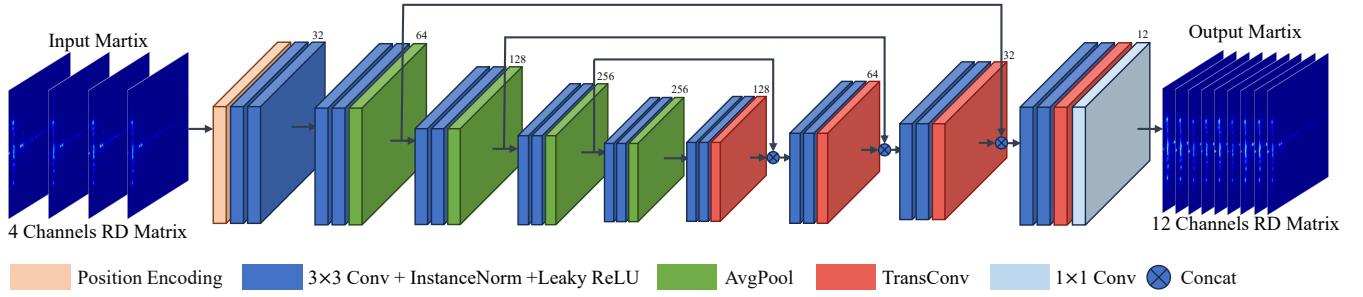


Fig. 7: Architecture of the proposed Radar Enhancement Module. It consists of a position encoding function and a U-Net-based network with skip connections.

$$f_{\text{ResNet-18}}(\mathbf{I}_{t=0}) = \mathbf{c}_{\text{vision}}^{\text{spatial}} \in \mathbb{R}^{512} \quad (9)$$

$$f_{\text{3DConv}}(\mathbf{I}) = \mathbf{c}_{\text{vision}}^{\text{temporal}} \in \mathbb{R}^{512} \quad (10)$$

where $\mathbf{I}_{t=0}$ is the first frame, and $\mathbf{I}$ is the entire image sequence.

Shared Decoder. To reconstruct spatial occupancy and temporal motion, we employ two decoders with a 5-layer ResNet architecture to handle spatial and temporal information (the detailed architecture is shown in Fig. 9).

To fuse complementary features from both the radar and the vision modalities, we use a shared occupancy decoder that learns from both spatial feature vectors. The primary function of this decoder is to predict whether a given point $\mathbf{p}(x, y, z) \in \mathbb{R}^3$ is occupied in the space. Specifically, for each modality $m \in \{\text{radar}, \text{vision}\}$ and each query point $\mathbf{p}(x, y, z) \in \mathbb{R}^3$, it predicts the occupancy probability, allowing us to model the spatial structure of the objects within it:

$$\hat{o}_{\theta_1}^m(\mathbf{p}(x, y, z)) = o_{\theta_1}(\mathbf{p}(x, y, z), \mathbf{c}_m^{\text{spatial}}). \quad (11)$$

Similarly, a shared velocity decoder learns from both temporal feature vectors. This decoder predicts the velocity of each point $\mathbf{p}(x, y, z, t) \in \mathbb{R}^4$, which enables us to model how these points move through space over time:

$$\hat{v}_{\theta_2}^m(\mathbf{p}(x, y, z, t)) = v_{\theta_2}(\mathbf{p}(x, y, z, t), \mathbf{c}_m^{\text{temporal}}). \quad (12)$$

Since the initial position $s(\mathbf{p}, 0) = \mathbf{p}$ can serve as the initial condition for a differential equation, the trajectory $s(\mathbf{p}, t)$ and the velocity predicted by the velocity network are constrained by the following relation:

$$\frac{\partial s}{\partial t}(\mathbf{p}, t) = \hat{v}_{\theta_2}^m(s(\mathbf{p}, t), t) \quad \text{with} \quad s(\mathbf{p}, 0) = \mathbf{p}. \quad (13)$$

We can solve this differential equation using a NODE solver [40]. This allows us to obtain the position of any point $\mathbf{p}(x, y, z)$ at any time t , ultimately generating a continuous trajectory $s(\mathbf{p}, t)$ of human motion in both space and time.

Contrastive Training. For the first frame of observation, we randomly sample a set of 3D query points $\mathbf{p}_0 \in \mathbf{P}_0$ from the first frame. The predicted occupancy values are compared with the ground truth using a Binary Cross-Entropy (BCE) loss. The loss is computed for each modality m , where m represents the different input modalities (e.g., radar, vision),

and the loss trains the parameters of the occupancy network (θ_1).

$$\begin{aligned} \mathcal{L}_{t_0}(\theta_1) &= \sum_m \mathcal{L}_{t_0}^m(\theta_1) \\ &= \sum_{\mathbf{p}_0 \in \mathbf{P}_0} \sum_m \text{L}_{\text{BCE}}(\hat{o}_{\theta_1}(\mathbf{p}_0, \mathbf{c}_m^{\text{spatial}}), o_{\text{gt}}(\mathbf{p}_0)) \end{aligned} \quad (14)$$

For a later time step $t \in (0, T]$, we aim to predict the correspondence of these points relative to different time steps using the velocity network. So we sample a set of 3D query points $\mathbf{p}_t \in \mathbf{P}_t$ and establish the correspondence by using the velocity network maps the points at time t back to their positions at time $t = 0$ under modality m , yielding $\hat{\mathbf{p}}_0^m$:

$$\hat{\mathbf{p}}_0^m = \mathbf{p}_t + \int_t^0 v_{\theta_2}(s(\mathbf{p}_t, \tau), \tau; \mathbf{c}_m^{\text{temporal}}) d\tau, \quad (15)$$

where $\frac{\partial s}{\partial \tau}(\mathbf{p}_t, \tau) = v_{\theta_2}(s(\mathbf{p}_t, \tau), \tau; \mathbf{c}_m^{\text{temporal}})$ and $s(\mathbf{p}_t, t) = \mathbf{p}_t$.

After mapping the points back to $t = 0$, we apply the occupancy network and compare the predicted occupancy with the ground truth occupancy at $t = 0$ using the BCE loss (Equation 16), which trains the parameters of both the occupancy network (θ_1) and the velocity network (θ_2).

$$\begin{aligned} \mathcal{L}_t(\theta_1, \theta_2) &= \sum_m \mathcal{L}_t^m(\theta_1, \theta_2) \\ &= \sum_{\mathbf{p}_t \in \mathbf{P}_t} \sum_m \text{L}_{\text{BCE}}(\hat{o}_{\theta_1}(\hat{\mathbf{p}}_0^m, \mathbf{c}_m^{\text{spatial}}), o_{\text{gt}}(\mathbf{p}_0)) \end{aligned} \quad (16)$$

During training, we use contrastive learning to enhance the model's ability to distinguish between similar and dissimilar features of radar and visual modality. As in CLIP [45], we construct sample pairs from the temporal and spatial features of both radar and visual data within the same batch and use cosine similarity to measure their similarity. For a batch of n samples, we have the following feature vectors:

$$\begin{aligned} \mathbf{C}_{\text{radar}}^{\text{spatial}} &\in \mathbb{R}^{n \times 512}, & \mathbf{C}_{\text{vision}}^{\text{spatial}} &\in \mathbb{R}^{n \times 512} \\ \mathbf{C}_{\text{radar}}^{\text{temporal}} &\in \mathbb{R}^{n \times 512}, & \mathbf{C}_{\text{vision}}^{\text{temporal}} &\in \mathbb{R}^{n \times 512} \end{aligned} \quad (17)$$

For spatial features, we construct sample pairs using $\mathbf{C}_{\text{radar}}^{\text{spatial}}$ and $\mathbf{C}_{\text{vision}}^{\text{spatial}}$, and measure their similarity using cosine similarity with a temperature parameter τ :

$$s_{ij} = \frac{\mathbf{C}_{\text{radar},i}^{\text{spatial}} \cdot \mathbf{C}_{\text{vision},j}^{\text{spatial}}}{\|\mathbf{C}_{\text{radar},i}^{\text{spatial}}\|_2 \|\mathbf{C}_{\text{vision},j}^{\text{spatial}}\|_2} \cdot \frac{1}{\tau}. \quad (18)$$

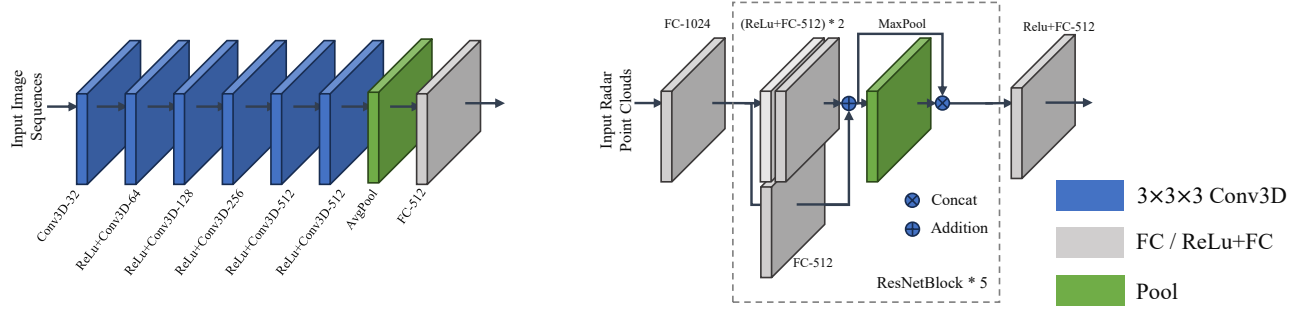


Fig. 8: Cross-Modal Encoder Architecture. Left: the vision temporal encoder, which processes the full image sequence; the vision spatial encoder uses ResNet-18 [44] and is omitted for space. Right: the radar temporal encoder, which takes the entire point cloud sequence as input and extracts temporal geometric features; the radar spatial encoder shares same network structure.

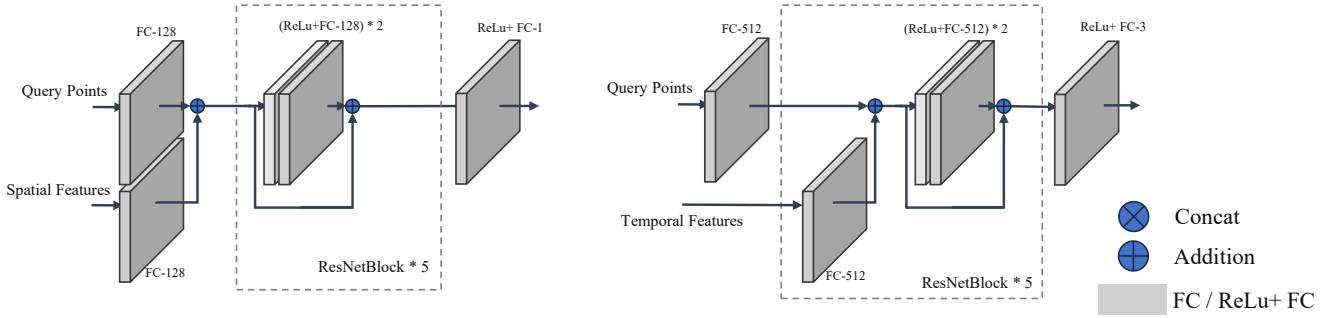


Fig. 9: Shared Decoder Architecture. Left: the spatial decoder takes the spatial feature vectors from both modalities and predicts occupancy for queried 3D point. Right: the temporal decoder takes the temporal feature vectors and predicts the velocity at each 4D spatio-temporal point (x, y, z, t) .

Within each mini-batch, feature pairs with the same index (i, i) are treated as positive pairs, while all pairs $(i, j), j \neq i$ serve as negative pairs. The spatial contrastive loss is defined as

$$\mathcal{L}_{\text{contrastive}}^{\text{spatial}} = -\frac{1}{n} \sum_{i=1}^n \log \frac{\exp(s_{ii})}{\sum_{j=1}^n \exp(s_{ij})}. \quad (19)$$

For the temporal features, we construct a similar contrastive loss function, and the total loss is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{t_0} + \mathcal{L}_t + \lambda (\mathcal{L}_{\text{contrastive}}^{\text{spatial}} + \mathcal{L}_{\text{contrastive}}^{\text{temporal}}) \quad (20)$$

4D Representation Inference. During inference, we first use the occupancy network to predict the probability distribution of the 3D human body in the initial frame. For the two modalities $m \in \{\text{vision}, \text{radar}\}$ and a spatial query point $\mathbf{p} = (x, y, z)$ at $t = 0$, the shared occupancy network models occupancy as a Bernoulli random variable and predicts a modality-specific probability $p_m(\mathbf{p})$. The associated Bernoulli variance is given by $\sigma_m^2(\mathbf{p}) = p_m(\mathbf{p})(1 - p_m(\mathbf{p}))$, which we use as a proxy to reflect the relative confidence of the modality-specific prediction. Under the assumption that the modality-specific probability estimates are approximately unbiased and that their variances correlate with prediction reliability, we combine them using an inverse-variance weighting scheme, i.e., $p(\mathbf{p}) = \frac{\sum_m \sigma_m^{-2}(\mathbf{p}) p_m(\mathbf{p})}{\sum_m \sigma_m^{-2}(\mathbf{p})}$, which down-weights less reliable predictions and can be viewed as a form of minimum-variance fusion.

We then employ the Multiresolution IsoSurface Extraction (MISE) [21] algorithm to extract the human shape. Next, we use the shared velocity network to predict the velocity of this point at time t . The velocity predictions from both modalities are fused using the same inverse-variance weighting scheme to obtain the final velocity estimate. As a result, we generate a continuous spatiotemporal model of human motion.

D. Activity Recognition

In this part, we will briefly introduce how we get the final activity recognition. Leveraging the spatio-temporal correspondence of the 4D implicit representation, we extract the corresponding vertices and use a 3-layer PointNet to obtain the final activity recognition.

For an image sequence $\mathbf{I} \in \mathbb{R}^{H \times W \times T \times 3}$ and point cloud sequences $\mathbf{P} \in \mathbb{R}^{N_p \times T \times 3}$, a 4D implicit representation can be reconstructed through the implicit fusion module, and can be extracted as the mesh sequences through the MISE algorithm which can be represented as:

$$\mathcal{M} = (V, F), \quad V \in \mathbb{R}^{N_v \times T \times 3}, \quad F \in \mathbb{Z}^{N_f \times T \times 3} \quad (21)$$

where $\mathcal{M}$ denotes the entire mesh sequence, consisting of a vertex sequence $V \in \mathbb{R}^{N_v \times T \times 3}$ and a face sequence $F \in \mathbb{Z}^{N_f \times T \times 3}$. N_v is the number of vertices per mesh, N_f is the number of faces per mesh, and T is the number of time steps,

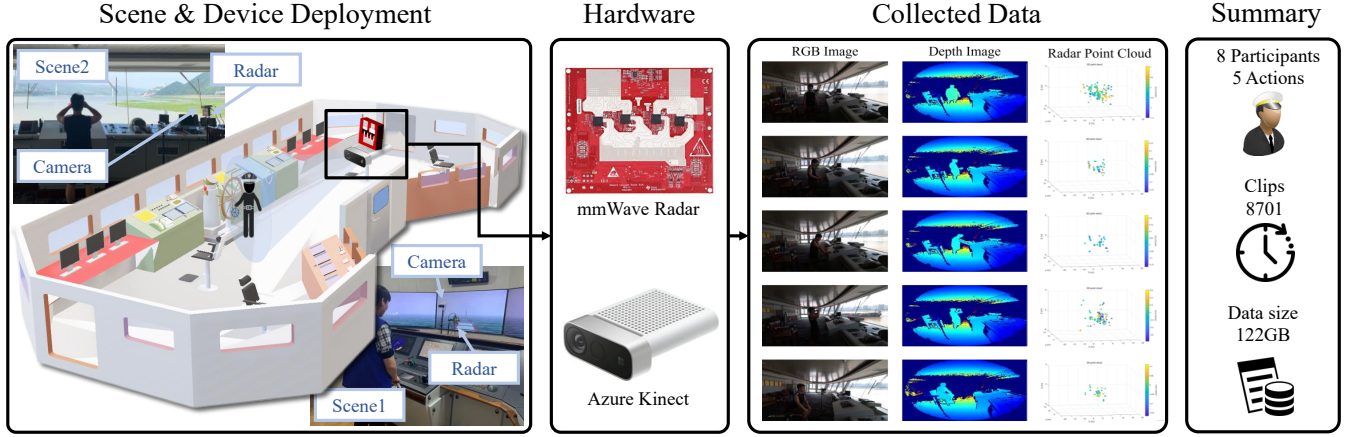


Fig. 10: Experimental setup for SIREN dataset. Data were collected in two different scenes using sensors from two modalities: vision and radar. A total of 8 volunteers participated, and five key activities of ship watch-keeping officers were defined. In total, 8701 clips were collected.

$\mathbb{Z}$ represents the integer indices defining the mesh faces. For the reconstructed mesh sequence of the human body over time, the vertex sequence V is extracted. A sliding window with a length of $L = 17$ and a stride of 1 is applied to divide V into overlapping subsequences, each of which can be represented as:

$$V^{(i)} = V[:, i : i + 17, :] \in \mathbb{R}^{N_v \times 17 \times 3} \quad (22)$$

where i is the starting index of the sliding window, and $V^{(i)}$ represents the i -th subsequence with a temporal length of 17. Each subsequence is then fed into a 3-layer PointNet with an input dimension of $N_v \times L \times 3$ to predict the activity class.

$$\hat{y}^{(i)} = \text{PointNet}(V^{(i)}), \quad \hat{y}^{(i)} \in \mathbb{R}^C \quad (23)$$

For training, we utilize a cross-entropy loss function to optimize the model parameters, defined as follows:

$$\mathcal{L}_{\text{recognition}} = - \sum_{j=1}^C y_j^{(i)} \log \hat{y}_j^{(i)} \quad (24)$$

where C is the total number of classes, $y_j^{(i)}$ represents the true label of sample i for class j , and $\hat{y}_j^{(i)}$ denotes the predicted probability of sample i belonging to class j .

IV. SHIP WATCH KEEPING OFFICER ACTIVITIES DATASET

Recognizing the activities of ship watch-keeping officers is critical for maritime safety. However, current datasets for this task are extremely scarce due to the following reasons: 1) most existing datasets are designed for land vehicle driver activity recognition and lack the unique characteristics of ship navigation scenarios; 2) these datasets often fail to capture complex environments such as those involving ship equipment and operational tasks. To address these gaps, we propose a new dataset by collecting visual and radar data from both simulated and real ship cabins. The dataset is open source and available here: SIREN_Dataset.

A. Data Collection Settings

We established a data acquisition platform to capture ship watch-keeping officer activities. For visual data acquisition, we utilized the Azure Kinect camera from Microsoft. The camera operates at 30 frames per second (FPS) with a resolution of 1280×720 pixels. Radar data were collected using the Texas Instruments IWR6843ISK single-chip mmWave radar combined with the DCA1000EVM data acquisition board and the MMWCAS-RF-EVM cascaded mmWave radar combined with the MMWCAS-DSP-EVM data acquisition board.

Data were collected in two different settings. The first setting involved a simulated ship cabin at the State Key Laboratory of Maritime Technology and Safety, Wuhan University of Technology, which provides a controlled environment for consistent observation under typical navigational contexts such as two ships meeting, entering port, and departing port. In addition, onboard data collection was conducted on two cruise ships, including the *Yangtze River Golden No.3* and the *Yangtze River Golden No.6*, capturing real-world watchkeeping activities throughout complete sailing processes. Fig. 10 illustrates the data collection settings. During data collection, we define five representative activity categories that cover the most common watchkeeping behaviors, including observing the surroundings with binoculars, steering, walking inside the cabin, sitting or maintaining posture, and interacting with equipment such as radar, control consoles, and VHF devices.

B. Data Calibration and Splits

To achieve time synchronization across different sensors, we employ the Network Time Protocol to perform time alignment. We prioritize the calibration of visual data and then partition the mmWave radar data based on the time index of both sensors. The processed dataset includes 1024 video sequences paired with corresponding radar data, which are further divided into 2-second intervals. Table IV summarizes the distribution of officer activities after segmentation. Further details are available on SIREN_Dataset.

TABLE IV: Distribution of officer activities.

Activities	Observing	Steering	Walking	Sitting	Operating	Total
Original Sequences	116	275	191	145	297	1024
Data Segmentation (2s)	377	3004	1471	1614	2235	8701

V. EXPERIMENTS & EVALUATION

In this section, we compare our model with existing activity recognition methods and test the effectiveness of the Radar Enhancement Module and the Implicit Fusion Module. We introduce the relevant datasets, baselines, implementation details, evaluation metrics, and the final results.

A. Datasets

mmBody Dataset [46]. This dataset contains RGBD images and real mmWave radar point clouds. The ground truth is represented as a 3D mesh of the human body using SMPL-X parameters. The dataset covers a variety of scenes, including laboratories, furnished rooms, rain, smoke, poor lighting, and occlusions. For activity recognition, we annotated different actions in the mmBody dataset, resulting in 8 meaningful activity classes.

SIREN Dataset [47]. The SIREN dataset is collected in maritime ship cabin environments and includes five key activities of ship watch-keeping officers across two types of cabin environments. It contains both radar and vision data, totaling 8701 clips, and is used for real-world recognition experiments in this work.

B. Baselines

We compare our method with representative activity recognition baselines covering radar-based, vision-based, and multimodal settings, including RadHAR [48] as a radar-based baseline, BPAI-Net [16] and MANet [49] as vision-based baselines, and MVFusion [50] as a multimodal radar-vision baseline. For the MVFusion baseline, we use its radar-vision fusion backbone to extract joint multimodal features, and apply the same 5-layer ResNet decoder to obtain the final activity recognition results.

C. Implementation Details

The main training configurations of the proposed method are summarized in Table V. For the Radar Enhancement Module, we set the number of training epochs to 100, the batch size to 16, and the learning rate to 3.141×10^{-4} . For the Implicit Fusion Module, the batch size is set to 16, the number of sampled points per input is 256, and the learning rate is 1×10^{-4} . The weight coefficient of the contrastive loss λ is set to 1×10^{-2} , and the temperature parameter τ is set to 0.1. In addition, a weight decay of 1×10^{-4} is applied to the velocity network to alleviate stiffness in the Neural ODE. For the activity recognition stage, we use a batch size of 16 and a learning rate of 1×10^{-4} during training. For inference, we set the occupancy threshold to 0.2 for 4D reconstruction to maximize IoU, and 0.4 for activity recognition to maximize accuracy. Most hyperparameters are selected based on standard practices in prior works and empirical tuning through

TABLE V: Training configuration of the proposed method.

Module	Parameter	Value
Radar Enhancement Module	Epochs	100
	Batch size	16
	Learning rate	3.141×10^{-4}
Implicit Fusion Module	Batch size	16
	Sampled points	256
	Learning rate	1×10^{-4}
	Contrastive weight λ	1×10^{-2}
	Weight decay	1×10^{-4}
Activity Recognition	Batch size	16
	Learning rate	1×10^{-4}

preliminary experiments. We further analyze the sensitivity of key parameters, including λ and the occupancy threshold, in Sec. VI-D.

D. Evaluation Metric

For evaluating the 4D reconstruction results, we use Intersection over Union (IoU) and Chamfer Distance (CD) as the metrics, averaged over all frames for sequences with multiple timestamps. For the final activity recognition results, we measure performance using Precision, Recall, and F_1 Score, weighted by the number of samples in each class. We also assess the performance of the Radar Enhancement Module by comparing the number of points in the radar point clouds before and after enhancement and use the Peak Signal-to-Noise Ratio (PSNR) to evaluate the quality of the range-angle maps.

E. Overall Performance

mmBody Dataset. On the mmBody dataset, we evaluate activity recognition performance under different sensing modalities, including fused, vision-only, and radar-only settings. The evaluation is conducted across five environmental conditions: Basic Scenes, Rain, Smoke, Poor Lighting, and Occlusion. The results are presented in Table VI. We further compare our method with representative activity recognition baselines on the mmBody dataset. The quantitative comparison results are reported in Table VIII.

SIREN Dataset. On the SIREN dataset, we evaluate our framework under different sensing modalities in real-world ship cabin environments. Experiments are conducted in both normal and low-light settings, comparing fused, vision-only, and radar-only configurations. The corresponding confusion matrices are shown in Fig. 17. Rows denote ground-truth labels and columns denote predicted labels, and activity classes are represented by abbreviations (Obs., Steer., Walk., Sit., Oper.). We also compare our method with representative activity recognition baselines. The quantitative comparison results are reported in Table VIII.

F. Comparison with Explicit Representations

To further validate the advantages of our implicit representation, we additionally include two commonly used explicit alternatives for comparison, i.e., skeleton-based and point

TABLE VI: Activity recognition results on the mmBody dataset across different environmental conditions.

Scene \ Sensor	Fused			Vision Only			Radar Only		
	Precision	Recall	F_1	Precision	Recall	F_1	Precision	Recall	F_1
Basic Scenes	93.03%	90.18%	0.92	81.65%	79.37%	0.80	74.62%	72.56%	0.73
Occlusion	83.76%	81.98%	0.82	56.37%	58.42%	0.57	65.48%	64.81%	0.65
Poor Lighting	82.18%	80.95%	0.81	52.51%	51.38%	0.52	68.65%	65.58%	0.67
Rain	82.64%	80.87%	0.81	68.69%	67.84%	0.68	63.47%	63.87%	0.63
Smoke	85.18%	83.63%	0.84	64.48%	63.42%	0.64	62.83%	61.61%	0.62

TABLE VII: Performance comparison of 4D reconstruction under different sensor configurations using the mmBody dataset.

Scene \ Sensor	Basic Scenes		Occlusion		Poor Lighting		Rain		Smoke	
	IoU	CD	IoU	CD	IoU	CD	IoU	CD	IoU	CD
Fused	72.65%	0.113	61.03%	0.176	65.62%	0.223	57.71%	0.208	62.91%	0.132
Vision Only	60.39%	0.181	46.35%	0.191	45.14%	0.288	51.13%	0.249	43.63%	0.240
Radar Only	44.66%	0.424	31.11%	0.546	41.70%	0.403	39.80%	0.612	38.01%	0.479

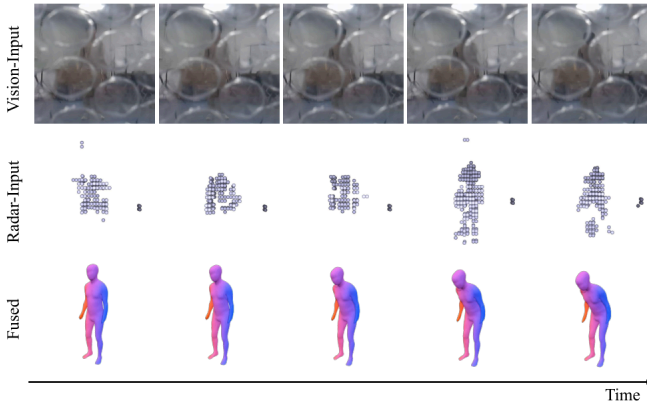


Fig. 11: 4D human reconstruction results for the bending motion sequence under the occlusion scenario, the colors indicate the correspondence across time.

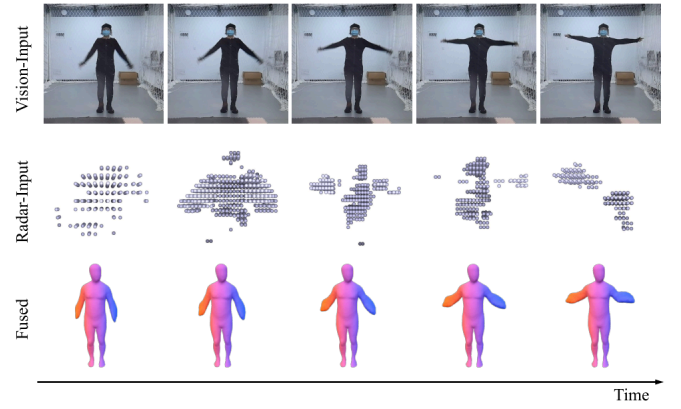


Fig. 12: 4D human reconstruction results for the raising both arms sideways motion sequence under the rain scenario, the colors indicate the correspondence across time.

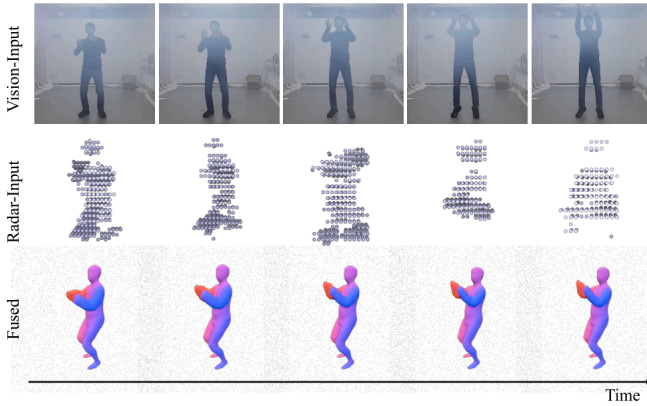


Fig. 13: 4D human reconstruction results for the raising both arms upward motion sequence under the smoke scenario, the colors indicate the correspondence across time.

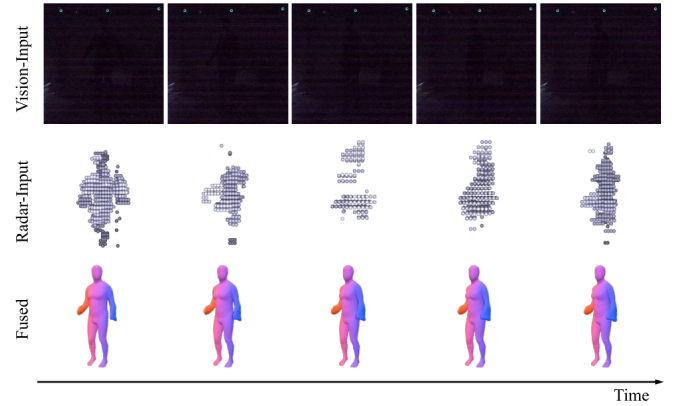


Fig. 14: 4D human reconstruction results for the twisting to the right motion sequence under the poor lighting scenario, the colors indicate the correspondence across time.

cloud-based representations. For the skeleton representation, we adopt the SMPL model [51] with 24 body joints; for the point cloud representation, we uniformly sample 1024 surface points on the human body. For both explicit representations, the encoder adopts the same network architecture as in Fig. 8.

The modality feature vectors are concatenated and fed into a spatial decoder in which the query-point branch is removed. In addition, we employ the same classifier for all methods to eliminate potential performance gaps caused by differences in downstream architectures. We present both quantitative and

TABLE VIII: Activity recognition comparison on the mmBody and SIREN datasets under different scene conditions.

Method	Modality	mmBody Dataset						SIREN Dataset					
		Basic Scenes			Adverse Scenes			Normal Scenes			Low-light Scenes		
		Precision	Recall	F_1	Precision	Recall	F_1	Precision	Recall	F_1	Precision	Recall	F_1
RadHAR	Radar	82.61%	80.07%	0.81	70.17%	68.92%	0.69	80.74%	79.61%	0.80	79.38%	78.49%	0.79
BPAL-Net	Vision	81.07%	83.20%	0.82	58.48%	61.45%	0.60	84.47%	84.72%	0.84	63.57%	64.92%	0.64
MANet	Vision	86.47%	87.51%	0.87	61.17%	60.75%	0.61	83.18%	81.69%	0.83	54.46%	57.81%	0.56
MVFusion	Vision+Radar	87.61%	84.18%	0.86	79.73%	78.98%	0.79	85.15%	84.78%	0.84	82.62%	82.41%	0.82
Ours	Vision+Radar	93.03%	90.18%	0.92	83.47%	81.95%	0.83	88.29%	86.58%	0.87	85.21%	83.20%	0.84

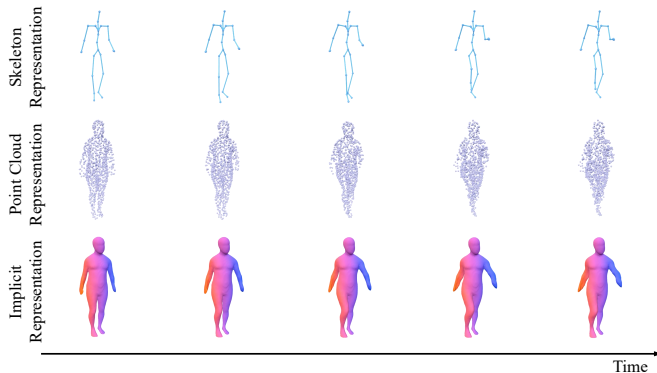


Fig. 15: Comparison of skeleton-based, point cloud-based, and implicit representations for a walking motion sequence.

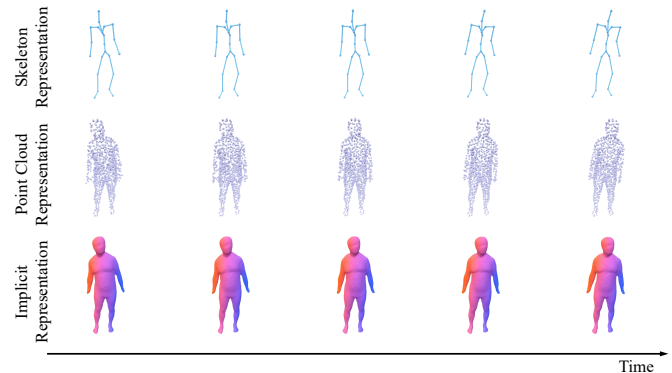


Fig. 16: Comparison of skeleton-based, point cloud-based, and implicit representations for a side-bending motion sequence.

TABLE IX: Comparison of different human representations for HAR under basic and adverse scenes on the mmBody dataset.

Representation	Basic Scenes			Adverse Scenes		
	Precision	Recall	F_1	Precision	Recall	F_1
Skeleton	81.26%	80.57%	0.81	69.95%	70.47%	0.70
Point Cloud	72.34%	70.73%	0.71	60.17%	59.60%	0.60
Ours	93.03%	90.18%	0.92	83.47%	81.95%	0.83

qualitative results in Table IX and Figs. 15–16, which show the advantages of the implicit representation over explicit alternatives under the same evaluation protocol.

G. Performance of Implicit Fusion Module

Quantitative Results. To compare the effectiveness and complementary advantages of different sensors in various environments, we evaluate the 4D reconstruction performance using the mmBody dataset under three configurations: vision-only, radar-only, and fused. Performance is measured using the IoU and CD metrics, and the results are summarized in Table VII.

Qualitative Results. We present qualitative results based on the mmBody dataset, as shown in Figs. 11, 12, 13, and 14. These figures illustrate different action sequences under four challenging scenarios: occlusion, rain, smoke, and poor lighting. The horizontal axis represents the timeline, and the colors indicate the correspondence across time. For each scenario, we show the visual input, point cloud input, and fused results to illustrate the benefits of multimodal fusion in challenging environments.

Probabilistic Quality and Calibration. We also evaluate whether the inverse-variance fusion strategy improves the

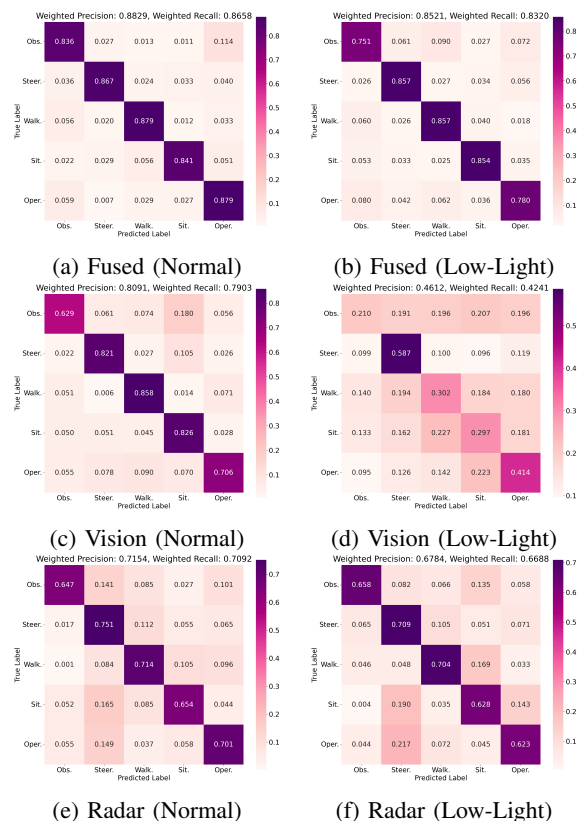


Fig. 17: Confusion matrices comparing fused, radar-only, and vision-only modalities in normal and low-light cabin environments.

TABLE X: Probabilistic quality and calibration comparison of different fusion strategies.

Method	Accuracy	Brier	ECE	MCE
Vision	0.998396	0.001414	0.001614	0.240443
Radar	0.997528	0.003036	0.009721	0.273378
Mean Fusion	0.998494	0.001675	0.005737	0.217358
IV Fusion	0.998494	0.001321	0.001616	0.187638

quality and calibration of the predicted probabilities. Given predicted probabilities p_i and binary labels $y_i \in \{0, 1\}$ for N samples, we report classification accuracy, Brier score, expected calibration error (ECE), and maximum calibration error (MCE). The classification accuracy is computed by thresholding the predicted probability at 0.5:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[(p_i \geq 0.5) = y_i]. \quad (25)$$

The Brier score measures the mean squared difference between predicted probabilities and binary labels:

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2. \quad (26)$$

For calibration evaluation, predictions are partitioned into $K = 15$ confidence bins $\{B_k\}_{k=1}^K$ according to their predicted probabilities. For each bin, the average confidence and empirical accuracy are defined as

$$\text{conf}(B_k) = \frac{1}{|B_k|} \sum_{i \in B_k} p_i, \quad \text{acc}(B_k) = \frac{1}{|B_k|} \sum_{i \in B_k} y_i. \quad (27)$$

Based on these bin-level quantities, ECE and MCE are computed as

$$\text{ECE} = \sum_{k=1}^K \frac{|B_k|}{N} |\text{acc}(B_k) - \text{conf}(B_k)|, \quad (28)$$

$$\text{MCE} = \max_{k \in \{1, \dots, K\}} |\text{acc}(B_k) - \text{conf}(B_k)|. \quad (29)$$

Thus, Accuracy measures the correctness of binary predictions after thresholding, the Brier score evaluates the overall probabilistic prediction quality, and ECE and MCE evaluate average and worst-case calibration gaps, respectively. The results are summarized in Table X.

H. Performance of Radar Enhancement Module

Quantitative Results. We evaluate the effectiveness of the Radar Enhancement Module by analyzing the density of radar point clouds generated before and after enhancement. As reported in Table XI, the average number of radar points per frame increases from 43.9 to 134.8 after enhancement. In addition to point cloud density, we assess the quality of the generated range-angle maps using PSNR. The enhanced range-angle maps achieve a mean PSNR of 41.53 dB with a standard deviation of 5.95, representing an average PSNR improvement of 13.0 dB over the range-angle maps.

Qualitative Results. Fig. 18 provides qualitative comparisons of the range-angle maps before and after applying the Radar

TABLE XI: Quantitative evaluation of the Radar Enhancement Module in terms of point cloud density and range-angle map quality.

Condition	Avg. Point Cloud Count	Mean PSNR (dB)	Std. (dB)
Before Enhancement	43.9	28.53	3.37
After Enhancement	134.8	41.53	5.95

Enhancement Module. After enhancement, the angular responses become sharper, with clearer target separation and reduced angular ambiguity.

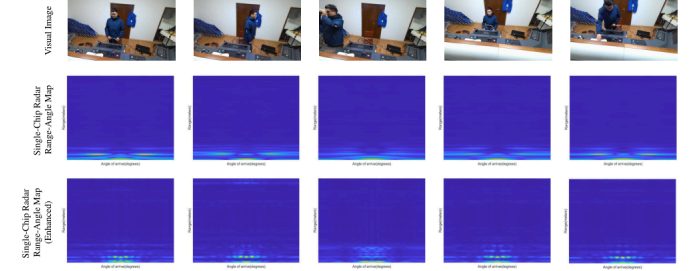


Fig. 18: We present the range-angle maps before and after the Radar Enhancement Module, along with their corresponding real images.

I. Adverse Scene Performance Analysis

To evaluate the robustness of different modalities under adverse scenes, we quantify how each modality degrades in each scene and how much performance gain is brought by our fused method.

We first define the *Scene Degradation Ratio* (DR) to measure the relative performance drop of the modality m in scene c :

$$\text{DR}_{m,c} = \frac{\text{IoU}_{m,c}}{\text{IoU}_{m,\text{Basic}}}, \quad (30)$$

where $\text{IoU}_{m,c}$ denotes the reconstruction IoU of the modality m in adverse scene c . A lower DR indicates stronger degradation relative to the basic scene.

We further compute the *Relative Sensor Advantage* (RSA) to assess how much the fused model improves over the best single modality in each adverse scene:

$$\text{RSA}_c = \text{IoU}_{\text{Fused},c} - \max(\text{IoU}_{\text{Vision},c}, \text{IoU}_{\text{Radar},c}). \quad (31)$$

The results are shown in Table XII. We can see that the vision modality degrades heavily in Occlusion, Poor Lighting and Smoke ($\text{DR} < 0.8$). In contrast, Radar remains comparatively stable in Poor Lighting, Rain, and Smoke ($\text{DR} > 0.85$), but shows a notable performance drop in Occlusion ($\text{DR} = 0.697$). The fused method remains relatively stable across all adverse scenes, with only a moderate drop in Rain ($\text{DR} = 0.794$). Its RSA values are positive in all adverse scenes, with large gains in Poor Lighting and Smoke, demonstrating the robustness benefits brought by multimodal fusion.

We also observe that incomplete reconstruction may occur under occlusion, as shown in Fig. 19. This happens when parts of the human body are not sufficiently observed.

TABLE XII: Scene Degradation Ratio (DR) and Relative Sensor Advantage (RSA) across adverse scenes.

Metric	Sensor	Adverse Scenes			
		Occlusion	Poor Lighting	Rain	Smoke
DR	Fused	0.840	0.903	0.794	0.866
	Vision	0.768	0.747	0.847	0.722
	Radar	0.697	0.934	0.891	0.851
RSA	-	+14.68	+20.48	+6.58	+19.28

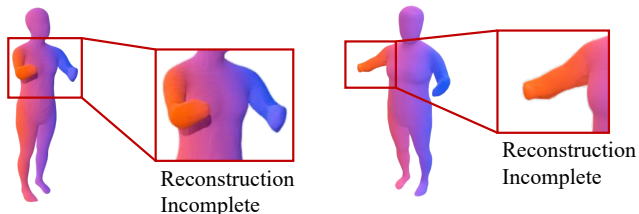


Fig. 19: Failure cases under occlusion, where incomplete reconstruction occurs due to limited observations.

J. Computational Cost Analysis

We compare the computational cost of our method with representative baselines in terms of the number of parameters and inference time. The results are shown in Table XIII.

Our model has 41.36M parameters, which is comparable to MANet and larger than MVFusion. The inference time is 94.23 ms per frame, which is higher than other methods due to the implicit representation and reconstruction process.

We further analyze the bottleneck and find that the main overhead comes from the mesh extraction using the MISE algorithm. This step requires iterative querying of the implicit field and dominates the inference time. In future work, we plan to directly utilize the implicit representation without explicit mesh extraction to improve efficiency.

VI. ABLATION STUDIES

In this section, we perform a series of ablation experiments to evaluate the effectiveness of the proposed modules. Specifically, we test different sampling strategies for the Radar Enhancement Module and investigate the impact of incorporating the Radar Enhancement Module (with or without) on recognition performance. We further compare the proposed Implicit Fusion Module with traditional fusion methods and validate the effectiveness of contrastive learning. In addition, we analyze the sensitivity of the model to key hyperparameters.

A. The Effectiveness of Radar Enhancement Module

We first test the results using different masking strategies for the Radar Enhancement Module. As shown in Fig. 20, the results for two different masking strategies are presented. We use PSNR as the evaluation metric and report the PSNR of the range-angle map in Table XIV. The results show that Masking Strategy 1, using channels R_0, R_4, R_8, R_{12} , achieves better performance.

TABLE XIII: Computational cost and performance comparison with baseline methods.

Model	Params (M)	Time (ms/frame)	Precision	Recall
RadHAR	0.15	1.19	82.61%	80.07%
BPAL-Net	33.51	3.64	81.07%	83.20%
MANet	43.24	11.19	86.47%	87.51%
MVFusion	15.43	46.57	87.61%	84.18%
Ours	41.36	94.23	93.03%	90.18%

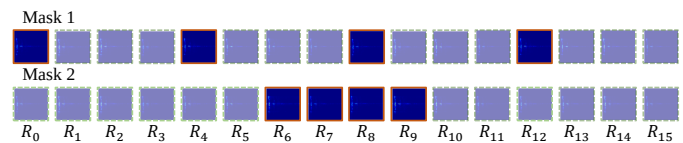


Fig. 20: Two different masking strategies.

TABLE XIV: PSNR comparison of range-angle maps under different masking strategies.

Masking Strategy	Mean PSNR (dB)	Std. (dB)
Mask Strategy 1	41.53	5.95
Mask Strategy 2	38.59	5.09

Next, we conducted tests on the SIREN dataset with and without the Radar Enhancement Module to assess its impact. The results of the experiments are shown in Table XV.

TABLE XV: Performance evaluation on the SIREN dataset with and without the Radar Enhancement Module.

Approach	Precision	Recall	F_1
w/ Radar Enhancement Module	88.29%	86.58%	0.87
w/o Radar Enhancement Module	85.29%	84.53%	0.85

B. The Effectiveness of Implicit Fusion Module

To evaluate the effectiveness of the proposed Implicit Fusion Module, we first examine the individual contributions of its key components, and then compare the module with conventional fusion mechanisms and Transformer-based fusion architectures.

1) *Ablation on Shared Decoder and Contrastive Learning:* We evaluate the individual effects of the shared decoder and contrastive learning within the Implicit Fusion Module. When the shared decoder is removed, two modality-specific decoders with the same architecture are used instead, without parameter sharing. Four configurations are evaluated: without either component, with the shared decoder only, with contrastive learning only, and with both components enabled. The results are reported in Table XVI.

2) *Comparison with Conventional Fusion Mechanisms:* We then compare the proposed implicit fusion module with commonly used fusion mechanisms, including feature addition and feature concatenation. The modality encoders and decoder are kept identical across all methods, and the only difference lies in the fusion operation applied to the radar and vision feature vectors.

In the *addition-based fusion* setting, the spatial and temporal feature vectors from the two modalities are combined

TABLE XVI: Ablation study on shared decoder and contrastive learning.

Shared Decoder	Contrastive Learning	IoU (%)	CD
Without	Without	65.90	0.218
With	Without	69.64	0.122
Without	With	67.27	0.159
With	With	72.65	0.113

via element-wise addition. In the *concatenation-based fusion* setting, the corresponding feature vectors are concatenated along the feature dimension before being fed into the decoder.

3) *Comparison with Transformer-based Fusion Architectures*: We further compare our implicit fusion module with Transformer-based fusion architectures, which are based on self-attention and cross-attention mechanisms. We replace the shared decoder with a Transformer-based decoder while keeping the modality encoders and input representations unchanged. Specifically, we adopt a four-layer Transformer decoder implemented using the standard `nn.TransformerDecoder` module, and train all variants under identical settings.

We evaluate three representative Transformer-based fusion configurations. In the *self-attention* setting, radar features, vision features, and queried points are concatenated into a unified feature vector sequence, which is used as both the query, key and value. In the *cross-attention with vision queries* setting, the concatenation of vision features and queried points serves as the query, while the concatenation of radar features and queried points serves as the key and value. In the *cross-attention with radar queries* setting, the roles are reversed, where the concatenation of radar features and queried points serves as the query, while the concatenation of vision features and queried points serves as the key and value.

Based on the fusion strategies described above, we next evaluate their performance on 4D reconstruction. The fusion methods are evaluated using the IoU and Chamfer Distance metrics. We further compare them with Vision-only and Radar-only baselines. Experiments were conducted on the mmBody dataset across Basic Scenes and Adverse Scenes (Rain, Smoke, Poor Lighting, Occlusion). The results are shown in Table XVII.

TABLE XVII: Comparison of different fusion strategies in basic and adverse scenes.

Approach	Basic Scenes		Adverse Scenes	
	IoU	CD	IoU	CD
<i>Conventional Fusion Mechanisms</i>				
Addition	67.88%	0.155	55.23%	0.223
Concatenation	69.17%	0.153	56.77%	0.223
<i>Transformer-based Fusion</i>				
Self-Attention	69.03%	0.147	59.84%	0.178
Cross-Attention (Vision Query)	65.37%	0.149	56.15%	0.192
Cross-Attention (Radar Query)	70.63%	0.127	61.42%	0.170
<i>Single-modality Baselines</i>				
Vision-Only	60.39%	0.181	47.74%	0.243
Radar-Only	44.66%	0.424	39.58%	0.528
Ours	72.65%	0.113	62.31%	0.170

C. The Effectiveness of Contrastive Learning

Contrastive learning is introduced in our framework to reduce feature misalignment between radar and vision, which is critical for effective multimodal fusion. To evaluate its impact, we conduct experiments with and without contrastive learning on the mmBody dataset. The performance comparison is shown in Table XVIII.

TABLE XVIII: Comparison with and without contrastive learning.

Approach	Basic Scenes		Adverse Scenes	
	IoU	CD	IoU	CD
w Contrastive Learning	72.65%	0.113	62.31%	0.170
w/o Contrastive Learning	69.64%	0.122	58.71%	0.183

We further quantify how contrastive learning reduces feature misalignment between radar and vision. We define a cross-modal misalignment ratio as

$$R_{\text{mis}} = \frac{S_{\text{misaligned}}}{S_{\text{aligned}}}, \quad (32)$$

which measures how strongly misaligned pairs resemble each other relative to aligned pairs. Here, S_{aligned} is computed as the average cosine similarity between paired radar-vision embeddings:

$$S_{\text{aligned}} = \frac{1}{N} \sum_{i=1}^N \frac{\langle f_i^{\text{radar}}, f_i^{\text{vision}} \rangle}{\|f_i^{\text{radar}}\|_2 \|f_i^{\text{vision}}\|_2}, \quad (33)$$

where f_i^{radar} and f_i^{vision} denote the latent features of the i -th sample from the radar and vision modalities, respectively. Similarly, $S_{\text{misaligned}}$ is the cosine similarity over all non-matching radar-vision pairs:

$$S_{\text{misaligned}} = \frac{1}{N(N-1)} \sum_{i=1}^N \sum_{\substack{j=1 \\ j \neq i}}^N \frac{\langle f_i^{\text{radar}}, f_j^{\text{vision}} \rangle}{\|f_i^{\text{radar}}\|_2 \|f_j^{\text{vision}}\|_2}. \quad (34)$$

A lower R_{mis} reflects better cross-modal consistency. As shown in Table XIX, contrastive learning reduces the misalignment ratio from 0.0626 to 0.0503, achieving a 19.65% relative reduction in R_{mis} .

TABLE XIX: Cross-modal similarity analysis with and without contrastive learning.

Approach	S_{aligned}	$S_{\text{misaligned}}$	R_{mis}
w Contrastive Learning	0.5105	0.0257	0.0503
w/o Contrastive Learning	0.5324	0.0334	0.0626

D. Hyperparameter Analysis

We conduct a sensitivity analysis to examine the robustness of the proposed framework with respect to hyperparameters in both training and inference stages.

First, we analyze the sensitivity of the model to the weight of the contrastive learning loss used during training. To this end, we vary the loss weight over a wide range on a logarithmic scale, from 0.001 to 3.3, while keeping all other training

settings fixed. During this analysis, we monitor the validation reconstruction error $\mathcal{L}_{\text{rec}}^{\text{val}} = \mathbb{E}_{\mathcal{D}_{\text{val}}} [L_{t_0}(\theta_1) + L_t(\theta_1, \theta_2)]$, where L_{t_0} and L_t denote the BCE occupancy reconstruction losses introduced in Eqs. 14 and 16.

Next, we analyze the sensitivity of the 4D representation inference to the occupancy threshold. We vary the threshold over a wide range from 0.1 to 0.9 while keeping all other settings fixed, and evaluate the 4D inference quality in terms of IoU as well as the final activity recognition accuracy. As shown in Fig. 21, the reconstruction error reaches its minimum at a contrastive learning weight of approximately 0.01. For the occupancy threshold analysis, mean IoU achieves its peak at 0.2, while recognition accuracy is maximized at 0.4.

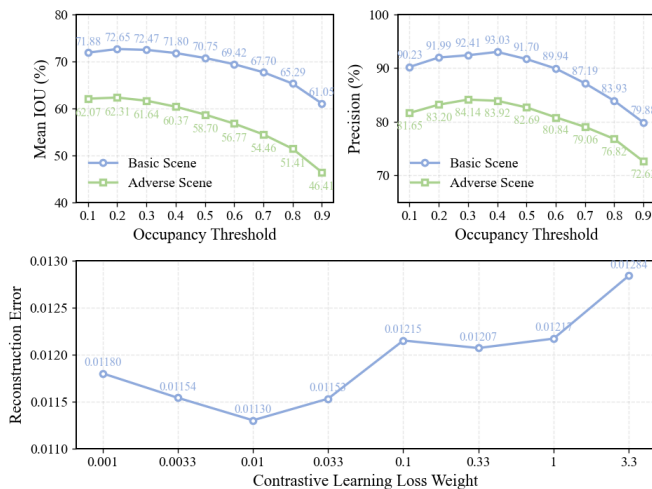


Fig. 21: Hyperparameter sensitivity analysis with respect to the contrastive loss weight and the occupancy threshold.

VII. CONCLUSION

In this paper, we presented a novel approach to ship watch-keeping officer activity recognition by using a 4D implicit representation to better integrate multimodal data. To bridge the resolution gap between radar and vision, we introduced the Radar Enhancement Module, which improves radar angular resolution and enhances the quality of multimodal fusion with camera data. Secondly, we leveraged contrastive learning within a shared neural network, enabling better cross-modal training. This approach mitigates the modality interference, capturing complementary features from radar and visual inputs, and improving the overall recognition performance. Lastly, we developed the SIREN dataset, specifically designed for ship cabin scenarios. The dataset consists of key activities performed by ship watch-keeping officers in both real and simulated environments. Experimental results demonstrate the effectiveness of our approach, showing a significant improvement over traditional multimodal fusion methods.

VIII. LIMITATIONS AND FUTURE WORK

While the proposed method demonstrates strong performance, there remain several directions for further improvement.

First, the implicit representation introduces relatively high computational cost due to the reconstruction process, and this has been analyzed in the computational cost evaluation. Future work could improve efficiency by avoiding the MISE-based mesh extraction and directly using the implicit representation for downstream tasks.

Second, under challenging conditions such as occlusion, improving the completeness of the reconstructed representation remains an important problem. Future work could enhance robustness by incorporating additional sensing modalities or introducing more detailed structural constraints.

Third, extending the model to more diverse environments and sensing configurations is another promising direction, including exploring broader data distributions and evaluating the method under more varied real-world conditions to further improve its generalization capability.

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